

Cache Translation Across Heterogeneous LLMs: An Audit with Identity Controls and Oracle Bounds

Abstract

Cache translation maps the key-value (KV) cache that a large teacher model forms over a context into the cache space of a smaller student model, so that the student answers without re-prefilling the context. A growing line of work reports quality-preserving translation across heterogeneous models. We audit this claim on Qwen3 pairs (4B→1.7B and 4B→0.6B) with a three-hypothesis framework and two instruments that prior evaluations omit. First, an *identity control* injects the student’s own cache through the full translation-and-injection pipeline: it reproduces the student’s own prefill exactly (per-sample logit cosine 1.000), which isolates mechanics from mapping. Second, *oracle probes* mix teacher-cache content into the student’s own cache and probe what the student can extract from translated content. Across six mapper families (per-head ridge, affine, per-layer affine, task-aware, a residual-anchored translator, and a per-head MLP), a controlled calibration ladder on a fixed evaluation set, and 42 recorded audit-protocol runs, no configuration makes the strong student exceed its own prefill: gold-probability change spans -0.14 to -0.47 against a self-kv control at $+0.000$. Calibration budget is inert (CHG -0.249 at $c=30$ vs -0.244 at $c=200$ on the same evaluation set), and nonlinearity does not help. The weak student reaches parity ($+0.010$, CI $[-0.08, +0.10]$). Oracle probes show a monotone decline as teacher content replaces student content in early and mid layers, robust to translator quality, and a perplexity–accuracy decoupling: a translator that restores near-native fluency (PPL 23.7 versus 21.2) still leaves accuracy at 0.267 versus 0.500. The teacher’s answer-relevant advantage, in short, does not survive KV-space translation under zero re-prefill: it sits in the teacher’s parameters, where the cache does not carry it. The audit protocol doubles as a checklist that a future cache-translation claim can run in one evaluation.

1 Introduction

Multi-model serving systems want to reuse the teacher’s work. A large teacher prefills a long context once; a smaller student should inherit that state and answer at teacher quality without re-reading the context. Cache translation formalizes this idea: learn a map $\mathbf{KV}^T \mapsto \widehat{\mathbf{KV}}^S$ from the teacher’s cache space into the student’s. Recent systems report encouraging results, including Mixture-of-Translators (MoT), which preserves 51.0% closed-set QA accuracy at 7B→0.5B scale [12], Cache-to-Cache semantic communication [5], latent-space communication [3], and selective KV sharing [16].

These reports conflate three questions that can fail independently. Does the injection pipeline consume a foreign cache correctly (*mechanics*)? Does the map replace the student’s own cache at replacement level (*mapping*)? Does the teacher’s cache carry student-readable content that lifts the student above its own prefill (*exploitability*)? Prior evaluations score the translated end-to-end pipeline against the student, so a positive delta could come from any of the three; a null result cannot be attributed. The line also lacks an identity control, a probe of the exploitability ceiling, and any report of what happens to *accuracy* when perplexity is repaired.

We evaluate cache translation as an audit with three gated hypotheses. **H1 (mechanics)** asks whether a foreign cache is consumed without loss; its instrument is an identity control that injects the student’s own cache through the full pipeline. **H2 (mapping)** asks whether any translator reaches replacement level; its instrument is a ladder of six mapper families under a disjoint calibration ladder of 30 to 200 examples, including a residual-anchored translator (RAT) that we derive from the two models’ own projections and a shared vocabulary alignment. **H3 (exploitability)** asks whether the teacher’s cache contains any student-readable advantage at all; its instrument is an oracle probe that mixes teacher-cache content into the student’s own cache at controlled fractions and layer windows. Figure 1 shows the gates and the instruments.

The audit returns three verdicts on Qwen3 pairs. H1 passes: the identity control reproduces the student’s own prefill with per-sample logit cosine 1.000 and a gold-probability change of +0.0001 (95% CI $[-0.0006, +0.0009]$), so mechanics is not the failure mode. H2 fails for the strong student: across the mapper ladder, the gold-probability change over the student’s own prefill spans -0.14 to -0.47 (best and worst seeds of the affine mapper; see Section 5), and the best single configuration reaches 0.365 against a student baseline of 0.503; the weak 0.6B student reaches parity (+0.010, CI $[-0.081, +0.102]$). H3 returns a principled negative: with 75% student cache and only 25% teacher content, accuracy already drops from 0.500 to 0.367, the decline is monotone in the teacher fraction, and replacing the top third of layers is the only harmless configuration. A translator that restores near-native fluency (PPL 23.7 versus 21.2 for the identity control) still leaves accuracy at 0.267. A cache can read fluently and still miss the answer; the deficit is not a fluency problem.

Three things fall out of the audit. The first is a framework: identity controls, disjoint calibration/evaluation splits, gold-probability metrics with bootstrap intervals and permutation tests, and oracle probes that bound exploitability without training anything (Section 4). The second is a bounded design space: six mapper families, a calibration ladder, component ablations, and a residual-anchored translator derived from the two models’ projections and a shared vocabulary, all under one protocol (Section 4, Section 5). The third is a verdict with consequences: mechanics passes, replacement fails for the strong student and holds for the weak one, exploitability is bounded at zero, and perplexity cannot stand in for accuracy as evidence (Section 5, Section 7).

2 A Survey of Cache-Level Communication Between Models

We organize prior work into four lines and close each with the assumption it leaves untested. Table 1 summarizes the comparison.

2.1 KV Reuse Within a Single Model

Prompt caching, cache-augmented generation, and streaming attention reuse a model’s own cache across requests. Streaming attention identifies attention sinks at early positions whose KV carries outsized attention mass [17]. CacheGen compresses and streams a KV cache across a network [13]; CacheBlend fuses several cached fragments with selective attention recomputation for RAG [19]; KVSharer shares caches across layers that behave dissimilarly [18]. These systems assume a single model on both sides of the cache. The reuse question is therefore about storage and attention cost, not about representation compatibility. Our audit inherits their serving motivation but moves the mismatch to the model axis, where the compatibility assumption breaks.

2.2 Cross-Model Cache Translation

The line closest to our audit translates caches between models. C2C proposes direct cache-to-cache semantic communication with a depth-ratio layer mapping [5]. LSC aligns caches through a shared latent space with a cross-attention translator [3]. MoT combines gated translator modules over depth-ratio channel windows with a Context Correction Loss that aligns a replayed target trajectory with the native one, and reports the strongest numbers to date: 51.0% average closed-set QA accuracy at Qwen2.5-7B→0.5B and 96.3% retention in long-context cache-augmented generation [12]. KVComm shares attention-ranked KV layers [16]; Interlat communicates last-hidden-state messages [4]; HCache restores caches from hidden-state translations [6]. Two properties of this line matter for our audit. First, its evaluations score the translated pipeline against the student baseline, which conflates mechanics, mapping, and exploitability. Second, its strongest variants do not operate under strict zero re-prefill: MoT reconstructs the target-side cache by replaying the context through the target model with source-guided sparse attention, so the reported quality includes a correction pass that a cold-start deployment would not get. Our audit holds the zero-re-prefill constraint fixed and asks what a translator alone can deliver.

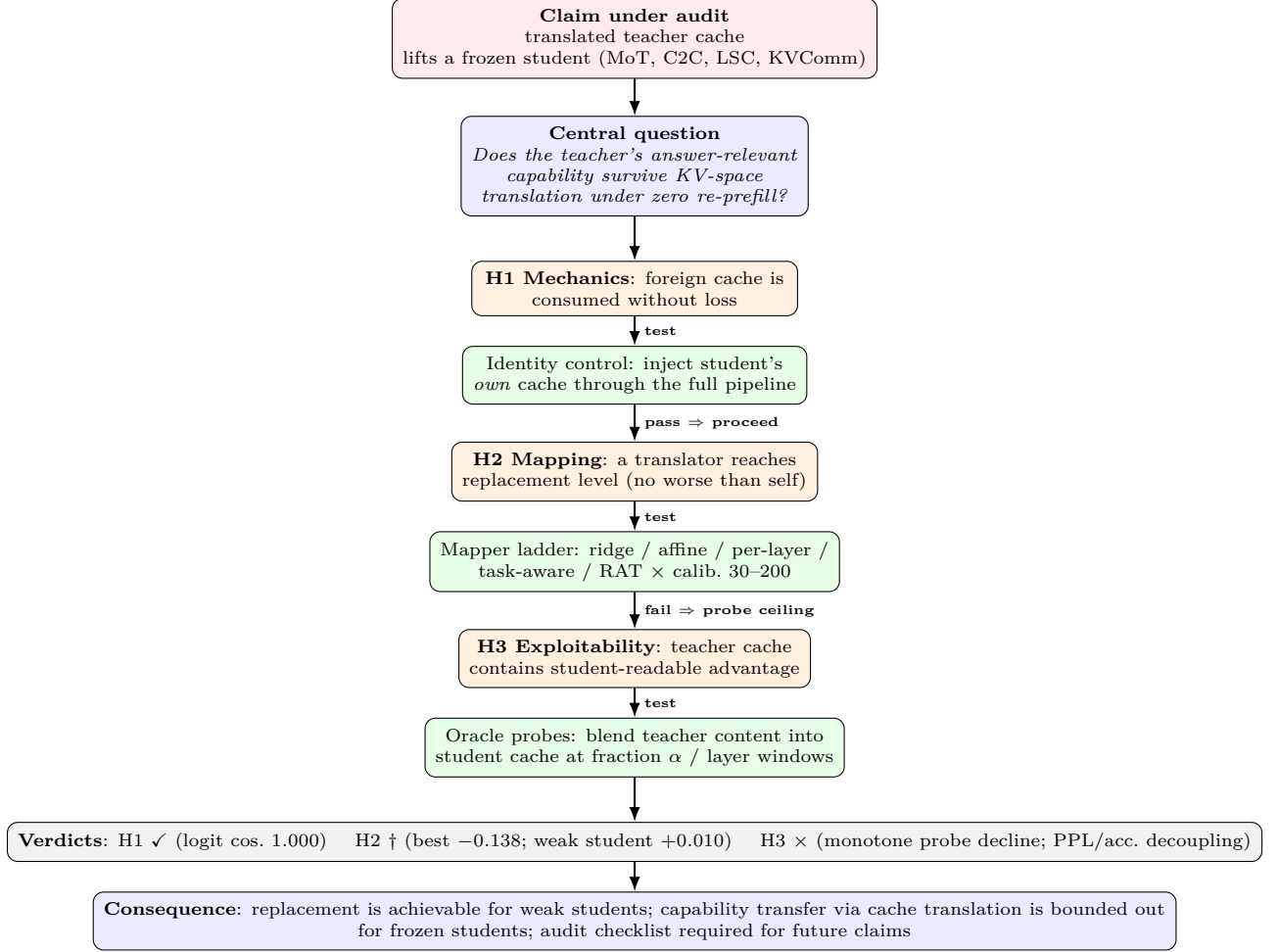


Figure 1: The three-hypothesis audit as a decision pipeline. Each hypothesis carries a dedicated instrument and a gate; H2’s failure routes the audit to H3’s oracle probes, which bound every translator without training one.

2.3 Cross-Model Representation Alignment

A separate line asks whether two trained networks’ representations can be aligned at all. Model stitching connects frozen networks with a learned linear layer and shows that stitching quality varies sharply across training runs and layers [1]. Relative representations replace absolute coordinates with anchor-based similarity spaces and enable zero-shot communication between encoders [14]. The Platonic Representation Hypothesis argues that representations converge across models up to linear transform [10], with refinements and proofs following [8, 20]. `vec2vec` translates embeddings between models without paired data by exploiting this shared geometry [11]. These results motivate the hypothesis that a linear map between cache spaces should exist. Our results qualify the conclusion for caches: the convergence evidence comes from pooled embedding spaces, while a cache is a per-layer, per-head, position-indexed state whose student-relevant content is not determined by its teacher counterpart (Section 6).

2.4 Frozen-Model Controllability

Work on steering shows that a frozen model’s behavior can be moved by injecting activations: task vectors compress in-context learning into a single mid-layer activation [9]; activation steering adds behavioral directions to the residual stream; gist tokens compress prompts into a few virtual tokens [15]. The tuned lens reads intermediate layers through per-layer affine probes [2], and the FFN-as- memory view assigns the factual

Table 1: Cache-translation and adjacent work, organized by what is translated and what is held fixed. “Replay” marks methods whose reported quality includes a target-side correction pass over the context; our audit holds strict zero re-prefill. None of the cited evaluations uses an identity control or an exploitability probe.

Work	What moves	Alignment	Replay	Unexamined assumption
C2C [5]	Full KV	depth-ratio lay- ers	no	a linear map per channel suffices
LSC [3]	KV via shared latent	learned latent space	no	one latent space fits all layers
MoT [12]	KV windows	gated translators	yes (sparse)	pipeline delta $\neq$ translation value
KVComm [16]	selected KV layers	attention ranking	no	ranked layers carry the signal
Interlat [4]	last hidden state	latent message	no	the last layer summarizes the cache
HCache [6]	hidden states	translator	no	states determine caches
This paper	full cache (zero re-prefill)	six families incl. RAT	no	<i>none</i> : identity control + oracle bounds

capacity of a model to its feed-forward weights [7]. This line supports the intuition that a frozen student can absorb injected information, which makes the H3 null result informative rather than tautological: the failure is not that frozen students cannot be steered. It is that the teacher’s cache never delivers the steering signal in a form the student can read. It also suggests where transferable signal might live: in residual-stream directions and mid-layer task vectors, not in KV entries.

2.5 What the Survey Does Not Establish

Across the four lines, no work (i) separates mechanics from mapping from exploitability, (ii) injects the student’s own cache as an identity control, (iii) reports accuracy at matched perplexity, or (iv) bounds exploitability with an oracle that requires no training. The claims of the translation line therefore rest on pipelines whose components have never been isolated. The audit below supplies the missing instruments.

3 The Three-Hypothesis Audit Framework

3.1 Setup

Let a teacher T prefill a context x and form a cache $C_T(x) = \{K_\ell^T, V_\ell^T\}_{\ell=1}^{L_T}$ with L_T layers, H KV heads, and head dimension d . A translator g_θ maps it to $\widehat{C}_S(x)$ in the space of a student S with L_S layers. The student then answers a query q with only the query tokens as input, $\hat{y} = S(q \mid \widehat{C}_S(x))$, under strict zero re-prefill: the context never enters the student’s forward pass. Three references anchor the audit. *Student self-prefill* $S(q \mid C_S(x))$ is the baseline the handoff must not degrade. *Identity injection* feeds $C_S(x)$ itself through the injection pipeline; it isolates mechanics from mapping. *Teacher full* $T(q \mid C_T(x))$ is the capability upper bound. We measure answer quality by the gold-letter probability p_{gold} , accuracy, and the change $\text{CHG} = p_{\text{gold}}(\text{handoff}) - p_{\text{gold}}(\text{student self})$, with 95% bootstrap intervals and sign-flip permutation tests; perplexity over the scoring suffix is recorded but never accepted as evidence of capability (Section 5.5).

3.2 Hypotheses, Instruments, Gates

Figure 1 states the three hypotheses with their instruments and gates. **H1**: the identity injection matches the student’s own prefill per sample (gate: mean logit cosine = 1 within float tolerance). **H2**: some g_θ reaches replacement level (gate: gold CHG ≥ 0 with the CI lower bound above a small tolerance $-\epsilon$; a stricter *gain* gate additionally requires the CI lower bound > 0). Replacement asks only that the student is not degraded; gain asks that the teacher’s cache provides a measurable lift. **H3**: the teacher’s cache contains

student-readable advantage (gate: some oracle probe configuration with teacher content improves over the student’s own cache). H3’s probe is decisive because it does not depend on training a translator; it directly measures what a frozen student can extract from teacher-cache content under the most favorable mixing.

4 Audit Methodology

4.1 Identity Control

At evaluation time we capture the student’s own cache over the context (allowed in calibration and probe stages), feed it through the same mapper interface, cache construction, position handling, and scoring path as a translated cache, and score. Any deviation from the student’s own prefill is mechanics loss. This is the control that prior evaluations omit.

4.2 Disjoint Splits, Unified Prompts, and Honest Counters

Mapper calibration and evaluation draws are disjoint samples of the same distribution (HellaSwag and ARC-Challenge, shuffled then split). All methods share one token-level prompt format, so the only difference between the handoff and the student baseline is where the context representation comes from. Zero-prefill counters read the actual cache length before and after the scoring forward pass rather than trusting bookkeeping, and the scoring pass uses an isolated cache because the deployed transformers version mutates a cache even under `use_cache=False`, which silently pollutes co-located measurements.

4.3 Mapper Ladder

We audit six families under one interface. **Ridge (per-head)**: a 128×128 ridge map per student layer and head with Gram aggregation over variable-length calibration samples. **Affine (per-head)**: the same with mean centering and an intercept, fit in closed form over concatenated calibration samples. **Affine (per-layer)**: heads merged into rows, eight times fewer parameters. **Task-aware**: ridge initialization plus a bounded task fine-tune that mixes reconstruction with gold-answer cross-entropy (weight $\beta=0.3$) under a diagonal parameterization with a per-layer bounded gate. **RAT (residual-anchored translator)**: an architecture-derived map

$$W^{(s,h)} = [\pi(W_V^{S,h}) R \pi(W_V^{T,h})]^\top, \quad R = \arg \min_R \|E_T R - E_S\|_F^2 + \lambda \|R\|_F^2, \quad (1)$$

where $W_V^{T,h}$ and $W_V^{S,h}$ are the two models’ own value projections at the aligned layer pair, $\pi(\cdot)$ is the pseudo-inverse pullback from a head’s value space to the residual stream, and R aligns the two residual streams through the shared 151,936-token embedding matrices, which both Qwen3 models ship. A closed-form rank- r correction fits the calibration residual on top of the analytic core, a Hungarian matching on head covariance profiles replaces the identity head correspondence, and the sink position (position 0) is overwritten with the student’s calibration statistics because its translated value is a PPL catastrophe (Section 5.3). Keys follow the same construction on W_K . Every family is evaluated at calibration budgets 30 and 200 with disjoint evaluation samples ($n=30$ and $n=63$ respectively).

4.4 Oracle Probes

The probes assume an oracle that no training can beat: they hand the student its *own* cache and mix in teacher content at a controlled budget. **Fraction probe**: $\alpha C_S + (1 - \alpha) \hat{C}_S$ for $\alpha \in \{0.25, 0.5, 0.75\}$ with $\alpha=0$ the student’s own cache. **Window probe**: translated content in only the bottom, middle, or top third of student layers, student content elsewhere. Because every configuration keeps the student’s own cache dominant or localized, any improvement over $\alpha=0$ would indicate exploitable teacher-cache content; monotone degradation under the tested translators rules out exploitable content in this regime. Probes are flagged non-deployable (they require the student’s own prefill) and exist only as measurement instruments.

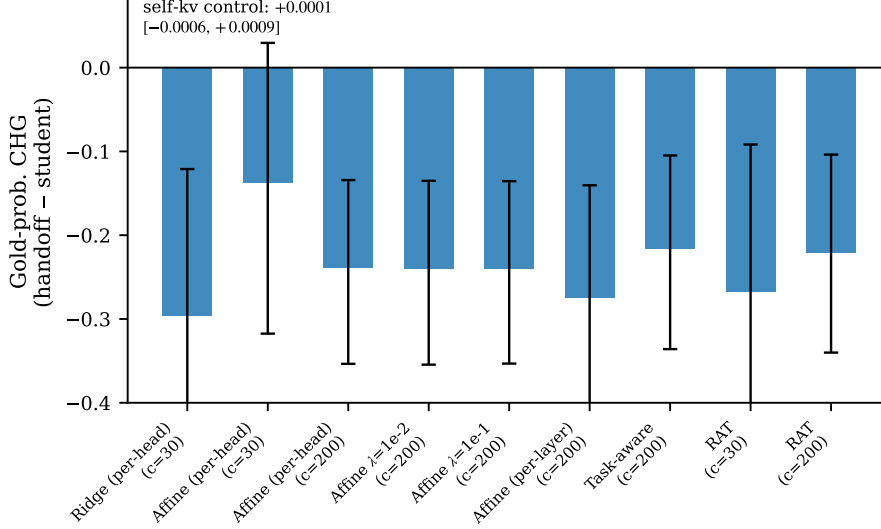


Figure 2: The mapper landscape for Qwen3-4B→1.7B under the unified audit protocol: gold-probability CHG over the student’s own prefill, with 95% bootstrap CIs. No family reaches zero; calibration budget moves affine mapping by only 0.005 on a fixed evaluation set; the architecture-anchored RAT and the nonlinear MLP sit inside the same band.

4.5 Pairs, Data, and Protocol Discipline

Primary pair: Qwen3-4B (36 layers, hidden 2560) → Qwen3-1.7B (28 layers, hidden 2048); contrast pair: → Qwen3-0.6B (hidden 1024). Both share the same tokenizer, vocabulary, $H=8$ KV heads, $d=128$, and RoPE base 10^6 . Every run records its git commit and protocol version in its artifacts; every number below is read directly from a recorded run.

5 Results

5.1 H1 Passes: Mechanics Is Not the Failure Mode

The identity control reproduces the student’s own prefill exactly: per-sample logit cosine 1.000, gold CHG +0.0001 $[-0.0006, +0.0009]$, $n=30$. The injection pipeline, cache construction, GQA head handling, and position accounting are therefore correct, and every subsequent gap is attributable to the translation itself. The same control verified the disk-persistence path: a cold-start evaluation that loads a persisted cache and mapper parameters reproduces the offline run bit-for-bit.

5.2 H2 Fails for the Strong Student and Holds for the Weak One

Figure 2 and Table 2 give the landscape. Centering is the single biggest gain: per-head ridge (-0.296) drops to per-head affine (-0.138), roughly halving the deficit. The teacher and student value distributions differ in location and scale before they differ in content, which is exactly what the measured norm ratios say ($1.58\times$ for keys, $7.44\times$ for values). The calibration budget, by contrast, buys nothing. Holding the evaluation set fixed ($n=100$) and varying only the calibration budget, affine mapping yields CHG -0.249 at $c=30$ and -0.244 at $c=200$, a difference of 0.005. The mapper has converged, and more data cannot close a gap that is set by the information content of the translated cache rather than by estimation variance. Structure and function class change little either: per-layer merging (-0.275), task-aware fine-tuning (-0.216), RAT (-0.221 to -0.268), and a per-head MLP mapper with two GELU layers (-0.303 at $c=30$, -0.236 at $c=200$) all sit inside the same band. Nonlinearity does not help. The strong student never approaches its own prefill (0.503); the best mapper reaches 0.365. The weak 0.6B student tells the opposite story. Affine at $c=200$ gives +0.010

Table 2: Main audit results. Acc/gold/PPL: answer accuracy, gold-letter probability, and perplexity of the scoring suffix for the translated cache (kv-both). CHG is the gold-probability change over the student’s own prefill with 95% bootstrap CIs. Pairs: Qwen3-4B→1.7B unless noted. All runs use disjoint calibration/evaluation splits and the unified prompt format.

Configuration	Acc / Gold / PPL	Gold CHG [95% CI]
Identity control (self-kv), 1.7B	0.500 / 0.503 / 21.2	+0.000 [−0.001, +0.001]
Student self-prefill, 1.7B	0.500 / 0.503 / 25.4	–
Teacher (full prefill), 1.7B	0.767 / 0.693 / –	–
Native average, 1.7B	0.167 / 0.169 / 536082.5	−0.334 [−0.508, −0.154]
Ridge per-head, c=30, 1.7B	0.200 / 0.207 / 88.5	−0.296 [−0.479, −0.121]
Affine per-head, c=30, 1.7B	0.367 / 0.365 / 46.1	−0.138 [−0.317, +0.029]
Affine per-head, c=200, 1.7B	0.254 / 0.258 / 56.4	−0.239 [−0.354, −0.134]
Affine per-layer, c=200, 1.7B	0.206 / 0.222 / 81441.1	−0.275 [−0.407, −0.140]
Task-aware, c=200, 1.7B	0.270 / 0.280 / 142.1	−0.216 [−0.336, −0.105]
RAT, c=30, 1.7B	0.233 / 0.235 / 501.6	−0.268 [−0.440, −0.092]
RAT, c=200, 1.7B	0.238 / 0.276 / 1008.2	−0.221 [−0.340, −0.104]
RAT no-anchor rank-32, c=30, 1.7B	0.267 / 0.280 / 23.7	−0.223 [−0.367, −0.088]
Affine per-head, c=200, 0.6B	0.302 / 0.283 / 71.8	+0.010 [−0.081, +0.102]
RAT, c=200, 0.6B	0.206 / 0.232 / 790.0	−0.041 [−0.120, +0.038]

[−0.081, +0.102], statistically indistinguishable from its own prefill, which is replacement-level service for a weak student.

5.3 RAT Component Ablations Separate Fluency From Capability

RAT ablations attribute the failure inside the architecture-anchored design. Removing the sink override explodes suffix perplexity to 2.7×10^5 : the translated position-0 entry, which carries outsized attention mass, is a catastrophe unless overwritten with the student’s own statistics. Removing the analytic core and raising the correction rank to 32 *repairs fluency*: PPL drops to 23.7, within 2.5 of the identity control. Accuracy stays at 0.267. The analytic core’s min-norm pullback is harmful on real models, and more importantly, fluency recovery does not touch the answer deficit. The cache can be made to look right to the language-modeling objective without becoming right for the task.

5.4 H3: Oracle Probes Bound Exploitability at Zero

The probes answer H3 without training anything (Figure 3). With 75% student cache and 25% teacher content, gold probability already falls from 0.5031 to 0.3798; the decay is monotone through $\alpha=0.75$. Window injection localizes the damage: replacing the top third of layers is indistinguishable from the student (0.5022 versus 0.5031), while the middle and bottom thirds are strongly harmful (0.2653, 0.2242). The pattern points to the architecture. Answer-relevant routing runs through early and mid layers, exactly where the teacher’s cache is least compatible; the top layers, where the translated content is least damaging, are not where the answer is decided. The probes were run with two translators: the RAT mapper reported above and the affine mapper (the best in the ladder, gold 0.365). Both show the same monotone decline. With affine at $\alpha=0.25$, gold CHG is −0.086 (CI [−0.18, +0.00]); at $\alpha=0.75$, −0.211 (CI [−0.41, −0.02]). The verdict is robust to translator quality: even the best available mapper produces content that the frozen student cannot exploit.

5.5 Perplexity and Accuracy Decouple

Figure 4 plots perplexity against accuracy for every method and run. The identity control and the student occupy the top-left regime. Translated caches populate a broad band, including a point with near-native perplexity and a 23-point accuracy deficit. Native injection sits at extreme perplexity (5.4×10^5) with low accuracy. Perplexity measures whether the cache permits fluent continuation; accuracy measures whether the

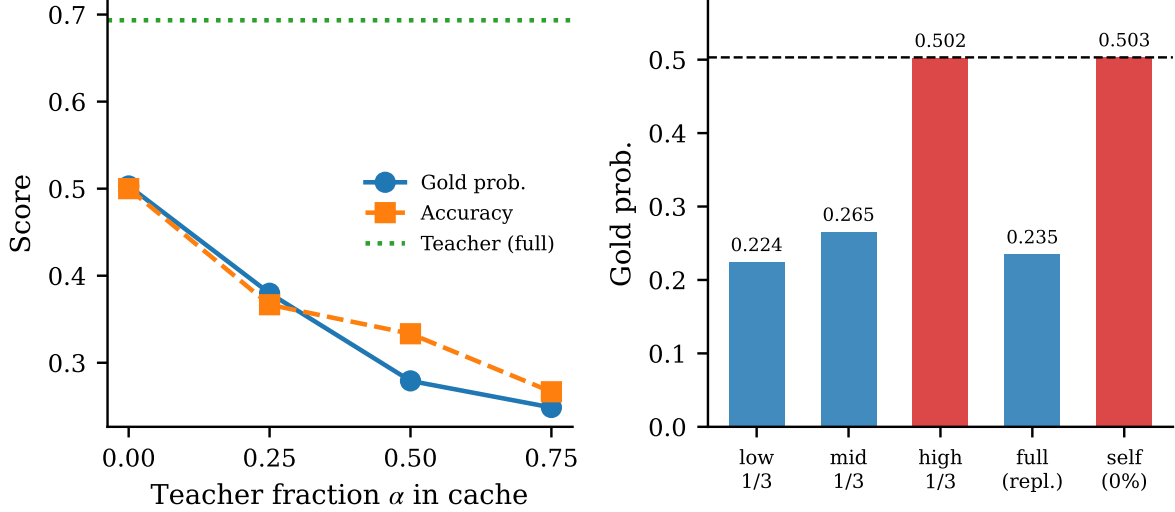


Figure 3: Oracle probes on Qwen3-4B→1.7B. Left: blending teacher-cache content into the student’s own cache degrades gold probability and accuracy monotonically in the teacher fraction α ; $\alpha=0$ is the student’s own cache. Right: window injection is harmless only in the top third of layers; full replacement and early/mid injection are far below the student.

cache supports the task. The two dissociate, so perplexity cannot serve as evidence of translation quality, a practice that appears in the serving literature because perplexity is cheap.

5.6 Channel Asymmetry

Channel ablations at $c=30$ rank value-only (0.367) above key-only (0.300) above both (0.200) on accuracy, while perplexity orders them in reverse (88 versus 253 versus 1.8×10^7). Mapping keys repairs fluency and damages routing: the mapped keys steer attention to plausible but task-irrelevant positions. We report the asymmetry as a finding and leave the mechanism open.

6 Why Translation Fails: An Architectural Analysis

The audit’s verdicts fit the architecture. A cache entry is a linear projection $W_V h$ of a residual-stream state h . Translation must recover h from V , move it across the residual-space gap (2560 to 2048), and re-project. Two facts bound the recovery. First, the pullback $\pi(W_V)V$ returns only the row-space component of h ; the component orthogonal to W_V ’s rows, which can dominate a 2560-dimensional state observed through a 128-dimensional head, is invisible to the teacher’s own cache and therefore unavailable to any translator, however nonlinear. The student’s corresponding value $W_V^S R h$ depends on that invisible component, so V_S is not a function of V_T : the mapping task is intrinsically underdetermined, and a learned translator can at best track the conditional mean. Second, the teacher’s advantage over the student sits in its parameters. FFN layers carry the model’s factual associations [7], and the capability gap between Qwen3-4B and 1.7B is a gap in these parameters, which a cache does not transport. The oracle probes make the consequence concrete: even with the student’s own cache holding 75% of the mass, the teacher’s 25% contribution degrades the answer. MoT’s own design corroborates the analysis: its reported quality includes a target-side replay with a correction loss, which is exactly the upper-layer correction that strict zero re-prefill denies. Under a cold start the translator’s output is all the student gets, and that output does not carry the teacher’s capability.

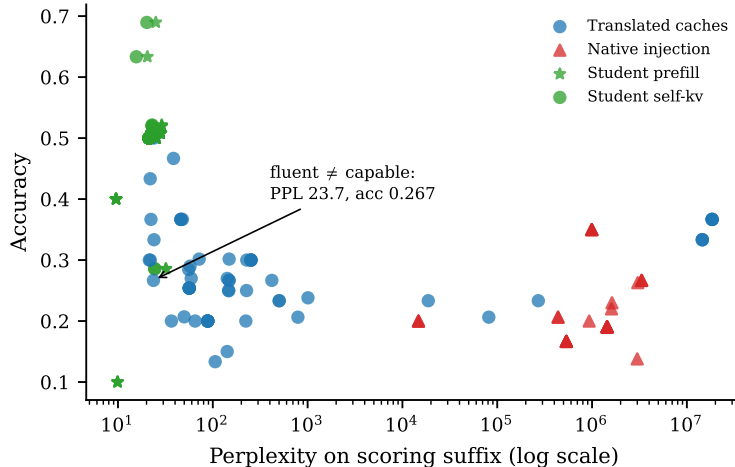


Figure 4: Perplexity over the scoring suffix versus accuracy across all methods and runs (log scale). The identity control and the student sit at low PPL and high accuracy; translated caches span two regimes; the best-reconstructed cache (PPL 23.7) remains 23 accuracy points below the student.

7 Implications for Cache-Translation Research

Any future cache-translation evaluation should ship three instruments: an identity control, a gold-probability metric with intervals, and an oracle probe. A pipeline delta over a student baseline is not evidence of translation, and a perplexity gain is not evidence of capability. Each instrument costs one evaluation run and composes with any serving stack.

For deployment, two regimes survive the audit. When the student is weak relative to the teacher (0.6B under 4B here), translated injection reaches parity with self-prefill, so a cached handoff saves the student’s prefill at no measured quality cost. Top-layer injection is harmless, which means a system that must spend translation capacity can spend it there, though our probes also show it buys nothing there. For strong students, cache translation is currently a net negative, and the honest fallback is the text channel: the teacher’s summary of the context scores 0.413 against the student’s 0.508 full-text baseline, so compression loses less than translation does on the strong student.

The survivable versions of the idea move work off the cache and onto the weights or the replay: allow a target-side correction pass the way MoT does, train student-side adapters, or translate through a shared latent space anchored by task vectors instead of per-entry caches. Each option gives up part of the cold-start guarantee, and each deserves a fresh identity control before its delta gets attributed to translation.

8 Limitations

Our audit covers one model family (Qwen3), two pairs, and two multiple-choice tasks with $n=30-63$ evaluation samples per run; the CI widths reflect that. The oracle probe bounds exploitability under linear blending and three-layer windows; a nonlinear exploit path outside this family would not be detected, though it would also need to survive the same measurement protocol. RAT’s analytic core assumes value projections are informative pullbacks of the residual stream; the synthetic validation confirms the mathematics, and the real-model failure is itself an audit finding. We did not re-implement MoT’s full replay pipeline; our claims concern the zero-re-prefill regime that its predecessors and our deployment target share. The mapper-ladder results use a single seed per configuration; a three-seed sweep of the best mapper (affine, $c=30$) confirms the negative sign across seeds 42, 43, and 44 (-0.14 , -0.47 , -0.46), with seed 42 being the most favorable case.

9 Conclusion

Cache translation promises to move a teacher’s work into a smaller model, and the audit separates that promise into three testable parts. The mechanism holds up: an identity control reproduces the student’s own prefill exactly. The map does not: six mapper families, a calibration ladder, and an architecture-anchored translator all leave the strong student below its own prefill and the weak student at parity. The map, moreover, is not where the problem sits. Oracle probes bound the student-readable advantage of the teacher’s cache at zero, and a near-native-perplexity cache still fails the task. The teacher’s capability is in its parameters. The audit framework is packaged so that the next cache-translation claim can be checked in a single run.

References

- [1] Yamini Bansal, Preetum Nakkiran, and Boaz Barak. Revisiting model stitching to compare neural representations. In *NeurIPS*, 2021. doi: 10.48550/arxiv.2106.07682.
- [2] Nora Belrose, Igor Ostrovsky, Lev McKinney, Zach Furman, Logan Smith, Danny Halawi, Stella Biderman, and Jacob Steinhardt. Eliciting latent predictions from transformers with the tuned lens. *arXiv preprint arXiv:2303.08112*, 2023.
- [3] L. M. Dery, Z. Yahav, H. Prior, Q. Feng, J. Shen, and A. Szlam. Latent space communication via KV cache alignment. *arXiv preprint arXiv:2601.06123*, 2026.
- [4] Xiao Du et al. Interlat: Latent-space message passing between language models. 2025.
- [5] Tianyu Fu, Zhen Min, Hang Zhang, Jun Yan, Guohao Dai, Wanli Ouyang, and Yu Wang. Cache-to-cache: Direct semantic communication between large language models. *arXiv preprint arXiv:2510.03215*, 2025.
- [6] Rui Gao et al. HCache: Cache restoration from hidden-state translation for efficient LLM serving. In *preprint*, 2025. Cited from MoT’s baseline description (MoT-h variant).
- [7] Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. Transformer feed-forward layers are key-value memories. In *Proceedings of EMNLP*, 2021. doi: 10.18653/v1/2021.emnlp-main.446.
- [8] Fabian Gröger, Shuo Wen, and Maria Brbić. Revisiting the platonic representation hypothesis: An aristotelian view. *arXiv preprint arXiv:2602.14486*, 2026.
- [9] Roei Hendel, Mor Geva, and Amir Globerson. In-context learning creates task vectors. In *Findings of EMNLP*, 2023. doi: 10.18653/v1/2023.findings-emnlp.624.
- [10] Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. The platonic representation hypothesis. In *arXiv preprint arXiv:2405.07987*, 2024.
- [11] Rishi Jha, Collin Zhang, Vitaly Shmatikov, and John X. Morris. Harnessing the universal geometry of embeddings. In *ICLR*, 2025. doi: 10.52202/085713-1530.
- [12] Jin-woo Lee, Minkyung Song, Junghyun Oh, Seunghoon Han, Soyoung Park, Gwangseon Jang, and Sungsu Lim. Mixture-of-translators: Translating KV caches across heterogeneous large language models. *arXiv preprint arXiv:2607.28979*, 2026.
- [13] Yuhan Liu, Hanchen Li, Yihua Cheng, Siddhant Ray, Yuyang Huang, Qizheng Zhang, Kuntai Du, Jiayi Yao, Shan Lu, Ganesh Ananthanarayanan, Michael Maire, Henry Hoffmann, Ari Holtzman, and Junchen Jiang. CacheGen: KV cache compression and streaming for fast large language model serving. In *Proceedings of the ACM SIGCOMM Conference*, 2024. doi: 10.1145/3651890.3672274.
- [14] Luca Moschella, Valentino Maiorca, Marco Fumero, Antonio Norelli, Francesco Locatello, and Emanuele Rodolà. Relative representations enable zero-shot latent space communication. *arXiv preprint arXiv:2209.15430 (ICLR 2023)*, 2022.

- [15] Jesse Mu, Xiang Lisa Li, and Noah D. Goodman. Learning to compress prompts with gist tokens. In *NeurIPS*, 2023. doi: 10.48550/arxiv.2304.08467.
- [16] Zheng Shi et al. KVComm: Selecting and sharing knowledge-rich KV cache layers between language models. 2025.
- [17] Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. In *arXiv preprint arXiv:2309.17453*, 2023.
- [18] Yifei Yang, Zouying Cao, Qiguang Chen, Libo Qin, Dongjie Yang, Hai Zhao, and Zhi Chen. KVSharer: Efficient inference via layer-wise dissimilar KV cache sharing. *arXiv preprint arXiv:2410.18517*, 2024.
- [19] Jiayi Yao, Hanchen Li, Yuhao Liu, Siddhant Ray, Yihua Cheng, Qizheng Zhang, Kuntai Du, Shan Lu, and Junchen Jiang. CacheBlend: Fast large language model serving for RAG with cached knowledge fusion. In *Proceedings of the ACM EuroSys Conference*, 2025. doi: 10.1145/3689031.3696098.
- [20] Liu Ziyin and Isaac L. Chuang. Proof of a perfect platonic representation hypothesis. *arXiv preprint arXiv:2507.01098*, 2025.

A Reproducibility

All numbers come from the `apcs` repository’s recorded runs (`reports/runs/*/inject-eval/`), each carrying its git commit, protocol version, configuration, and per-sample scores. The audit protocol is implemented in `apcs.inference.evaluator` with the flags `calib_eval_split`, `calib_shuffle`, `probe_mode`, and `native_rescale`; mapper families live in `apcs.mapper` (`math`, `joint`, `task_aware`, `rat`); persistence and the cold-start online phase in `apcs.inference.kv_store`. Unit tests cover the audit counters, the store round-trip, and the mapper interface.

B Additional Measurements

The teacher/student KV norm ratios are $1.58\times$ (keys) and $7.44\times$ (values), measured over 30 calibration contexts per layer. Norm rescaling of the native cache repairs neither its perplexity (5.4×10^5) nor its accuracy, ruling out scale as the primary failure. The unified prompt renders choices once; the earlier protocol rendered them twice for the student baseline, which inflated the confidence-based deltas reported in our own preliminary sweeps and motivated the unified format in the audit.